Why OCR Accuracy Alone Can Be Misleading

ARTIFICIAL INTELLIGENCE · MACHINE LEARNING · DEEP LEARNING

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SWIPE ->

Imagine an OCR system that correctly reads 98 out of every 100 characters.

That sounds excellent.

But what happens when the two incorrect characters are in a bank account number, a medical identifier, or an invoice total?

The impact can be much greater than the average accuracy suggests.

This is why OCR evaluation needs more than one metric.

Depending on the application, I would look at:

- Character Error Rate (CER)
- Word Error Rate (WER)
- Exact-match accuracy for critical fields
- Field-level precision and recall
- End-to-end document extraction accuracy

CER measures character-level errors relative to the reference text.

WER measures word-level errors.

Exact-match accuracy asks whether the entire field is correct.

These metrics answer different questions.

A system might have a low CER but still struggle with exact-match accuracy on long identifiers.

And a document pipeline might recognize text correctly but extract the wrong value because its layout analysis failed.

My takeaway:

Measure the failure that matters to the user, not just the metric that looks best on a dashboard.

#OCR #MLOps #AIEngineering

Thanks for reading.

Follow **Yasir Usman** for more on AI, ML and computer vision.

Code: github.com/yasirusman85/Tank